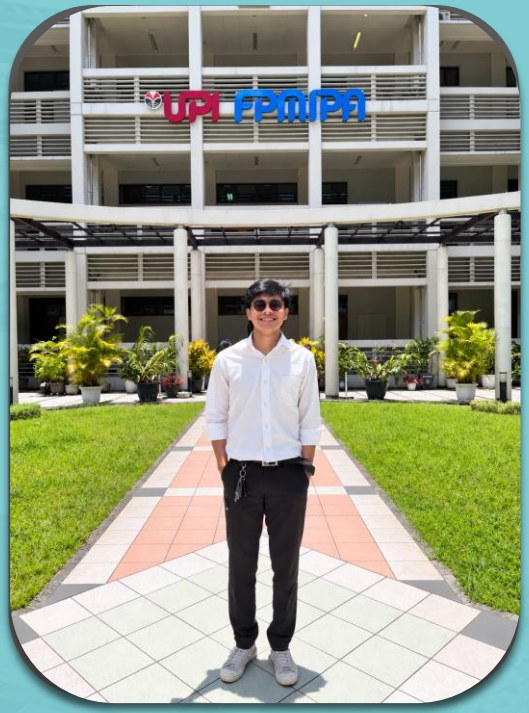
Rakamin Academy X kimia farma

Performance Analytics Kimia Farma

Kimia Farma - Big Data Analytics

Presented by
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Rakamin Academy

Hello, my name is..

Muhammad **Alfa**ruqi Ramzan Mahrudin

I'm an

Aspiring Data Analyst

Mathematics graduate from Universitas Pendidikan Indonesia, currently gaining hands-on experience as a **Data Analyst Support Intern** at **PT Cemerlang Statistika Indonesia (StatsMe)**. I also have experience in data verification and document management at **OJK** and the **Ministry of Finance**. Familiar with **SQL, Tableau, Python, Microsoft Office, and Google Workspace** for data processing and visualization.

Courses and Certification



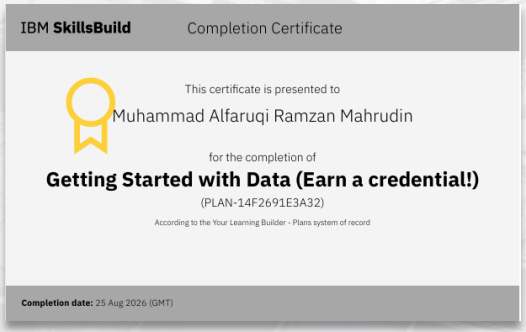
Oct – Dec 2025



Sep 2026



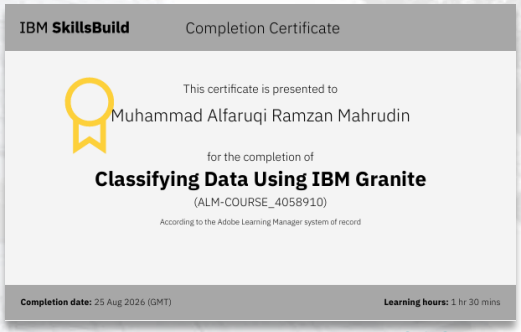
Sep 2026



Aug 2026



Aug 2026



Aug 2026

About Company

- **1817 -** Established as one of Indonesia's earliest pharmaceutical companies
- **2001 -** Became a publicly listed company
- **2020–2026 -** Joined the Bio Farma pharmaceutical holding and later officially changed its name to PT Kimia Farma (Persero) Tbk

Project Portfolio

Background

Evaluate Kimia Farma's business performance from 2020 to 2023 by analyzing **sales**, **profit**, **transactions**, **regional performance**, and **branch performance**.

Data Available

kf_final_transaction.csv

- Transaction_id
- Date
- Branch_id
- Customer_name
- Product_id
- Price
- Discount_percentage
- Rating

kf_inventory.csv

- Inventory_ID
- Branch_id
- Product_id
- Product_name
- Opname_stock

kf_kantor_cabang.csv

- Branch_id
- Branch_category
- Branch_name
- Kota
- Provinsi
- Rating

kf_product.csv

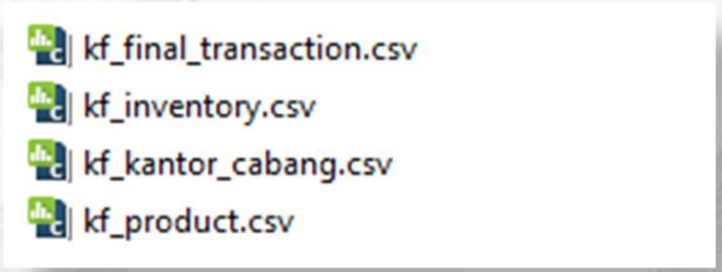
- Product_id
- Product_name
- Product_category
- Price

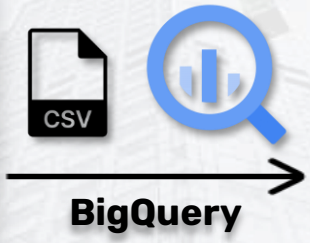
Problem Statement

How can the available data be used to evaluate sales, profitability, regional performance, and branch performance?

Project explanation video: Here!

1. Importing Dataset to BigQuery







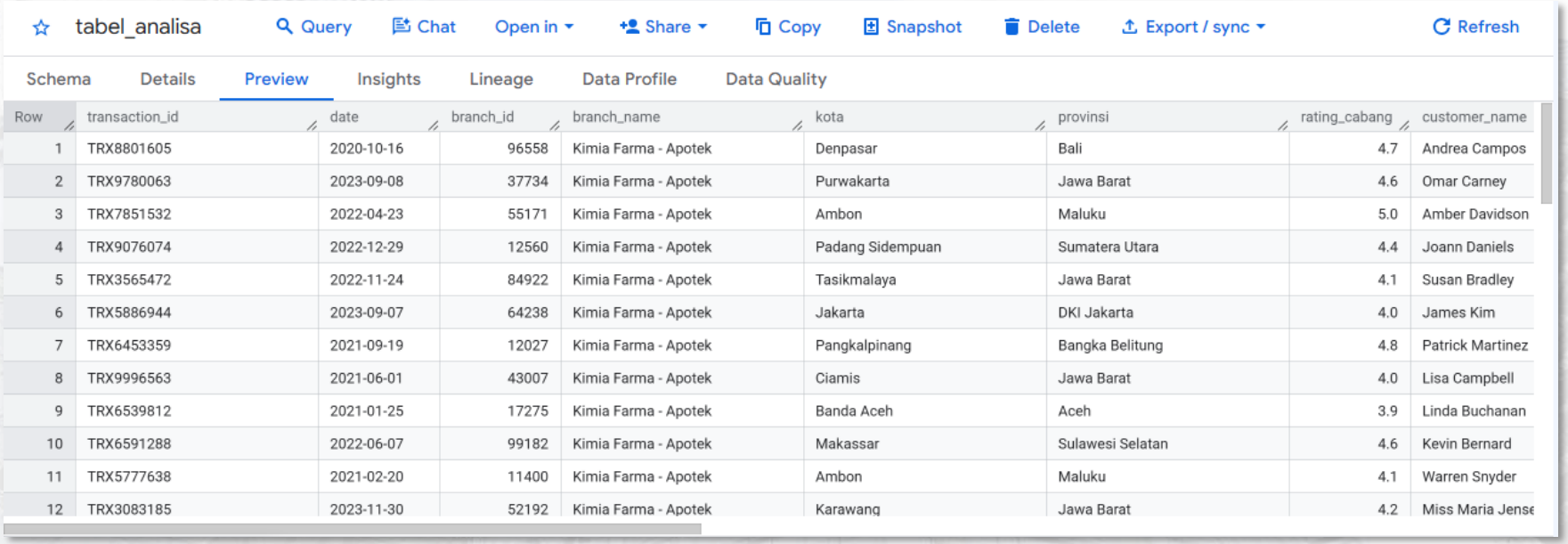
The project data was initially available as four **.csv** files containing **transaction**, **inventory**, **branch**, and **product data**.

Each CSV file was **uploaded** as a separate table into the kimia_farma dataset, keeping the original source names without the .csv extension. The imported tables were then checked and prepared for further **analysis**.

Role of **BigQuery**

Centralize multiple source datasets | Perform **SQL-based** transformation | Connect processed data directly to **Looker Studio**

2. Tabel Analisa



tabel_analisa | Query | Chat | Open in | Share | Copy | Snapshot | Delete | Export / sync | Refresh

Schema | Details | Preview | Insights | Lineage | Data Profile | Data Quality

Row	transaction_id	date	branch_id	branch_name	kota	provinsi	rating_cabang	customer_name
1	TRX8801605	2020-10-16	96558	Kimia Farma - Apotek	Denpasar	Bali	4.7	Andrea Campos
2	TRX9780063	2023-09-08	37734	Kimia Farma - Apotek	Purwakarta	Jawa Barat	4.6	Omar Carney
3	TRX7851532	2022-04-23	55171	Kimia Farma - Apotek	Ambon	Maluku	5.0	Amber Davidson
4	TRX9076074	2022-12-29	12560	Kimia Farma - Apotek	Padang Sidempuan	Sumatera Utara	4.4	Joann Daniels
5	TRX3565472	2022-11-24	84922	Kimia Farma - Apotek	Tasikmalaya	Jawa Barat	4.1	Susan Bradley
6	TRX5886944	2023-09-07	64238	Kimia Farma - Apotek	Jakarta	DKI Jakarta	4.0	James Kim
7	TRX6453359	2021-09-19	12027	Kimia Farma - Apotek	Pangkalpinang	Bangka Belitung	4.8	Patrick Martinez
8	TRX9996563	2021-06-01	43007	Kimia Farma - Apotek	Ciamis	Jawa Barat	4.0	Lisa Campbell
9	TRX6539812	2021-01-25	17275	Kimia Farma - Apotek	Banda Aceh	Aceh	3.9	Linda Buchanan
10	TRX6591288	2022-06-07	99182	Kimia Farma - Apotek	Makassar	Sulawesi Selatan	4.6	Kevin Bernard
11	TRX5777638	2021-02-20	11400	Kimia Farma - Apotek	Ambon	Maluku	4.1	Warren Snyder
12	TRX3083185	2023-11-30	52192	Kimia Farma - Apotek	Karawang	Jawa Barat	4.2	Miss Maria Jense

A single analytical table was created to **combine** the fields required for **business performance analysis**.

Fields:

TRANSACTION
- transaction_id
- date
- customer_name
- discount_percentage
- rating_transaksi

BRANCH
- branch_id
- branch_name
- kota
- Provinsi
- rating_cabang

PRODUCT
- product_id
- product_name
- actual_price

CALCULATED FIELDS
- persentase_gross_laba
- nett_sales
- nett_profit

3. BigQuery Syntax

The analytical table was created using **SQL** by joining **transaction**, **branch**, and **product** data, then calculating the required **sales** and **profitability** metrics.

Build & Join Tables

```sql
-- analytical table for the dashboard
CREATE OR REPLACE TABLE kimia_farma.tabel_analisa AS
SELECT
  a.transaction_id,
  a.date,
  c.branch_id,
  c.branch_name,
  c.kota,
  c.provinsi,
  c.rating AS rating_cabang,
  a.customer_name,
  d.product_id,
  d.product_name,
  d.price AS actual_price,
  a.discount_percentage,

```

Transaction data is connected to branch data using **branch_id** and product data using **product_id**.

Gross Profit Logic

```sql
    -- gross profit percentage based on product price
    CASE
      WHEN d.price <= 50000 THEN 0.10
      WHEN d.price > 50000 AND d.price <= 100000 THEN 0.15
      WHEN d.price > 100000 AND d.price <= 300000 THEN 0.20
      WHEN d.price > 300000 AND d.price <= 500000 THEN 0.25
      WHEN d.price > 500000 THEN 0.30
    END AS persentase_gross_laba,

    -- nett sales after discount
    ROUND(d.price * (1 - a.discount_percentage), 0) AS nett_sales,

```

Gross profit percentage is assigned based on the product **price range**, from **10%** to **30%**.

Sales & Profit Calculation

```sql
    -- nett profit after gross profit and discount
    ROUND(
      (
        d.price *
        CASE
          WHEN d.price <= 50000 THEN 0.10
          WHEN d.price > 50000 AND d.price <= 100000 THEN 0.15
          WHEN d.price > 100000 AND d.price <= 300000 THEN 0.20
          WHEN d.price > 300000 AND d.price <= 500000 THEN 0.25
          WHEN d.price > 500000 THEN 0.30
        END
      )
      - (d.price * a.discount_percentage),
      0
    ) AS nett_profit,

    a.rating AS rating_transaksi

  FROM kimia_farma.kf_final_transaction a
  JOIN kimia_farma.kf_kantor_cabang c
    ON a.branch_id = c.branch_id
  JOIN kimia_farma.kf_product d
    ON a.product_id = d.product_id;
```

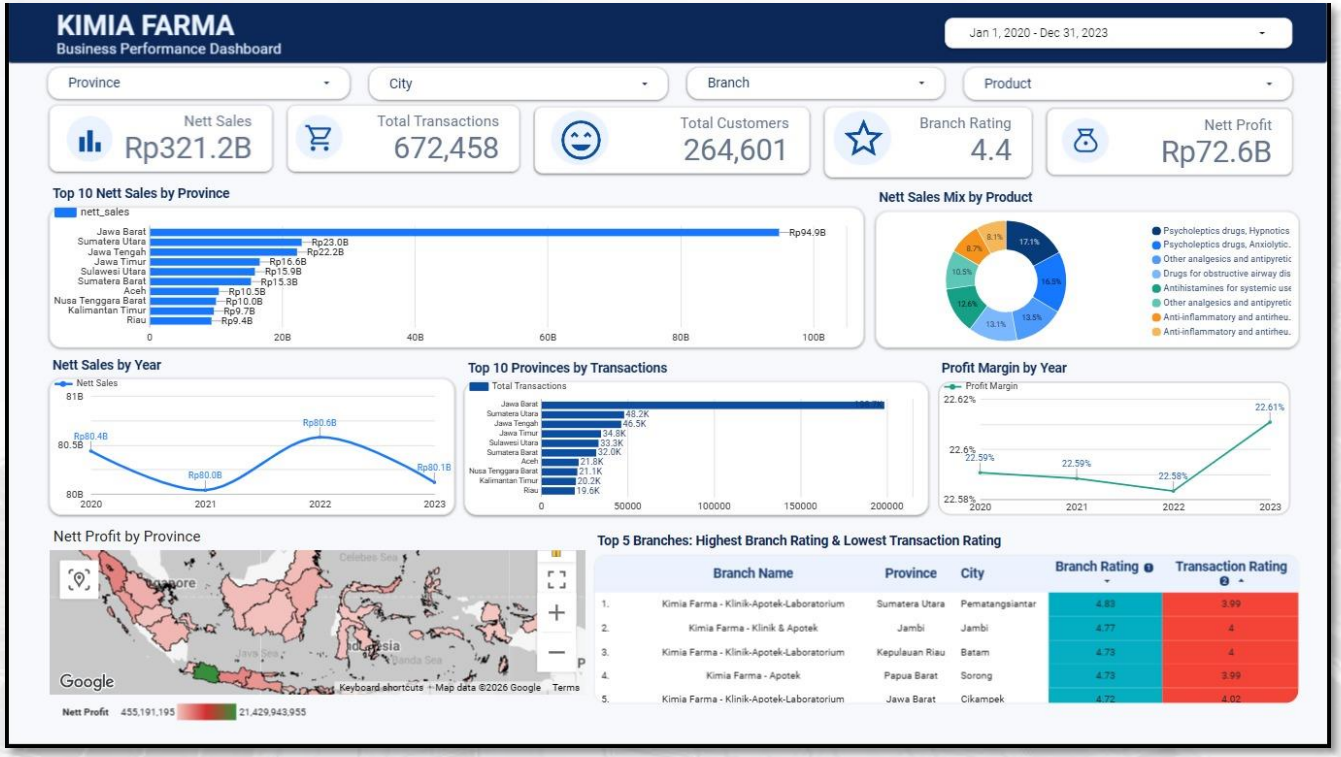
Nett Sales : Actual Price × (1 − Discount)
Nett Profit : (Actual Price × Gross Profit %) − (Actual Price × Discount %)

SQL Repository: tabel_analisa.sql

4. Dashboard Performance Analytics

Dashboard Design & Analysis Logic

- **Single-page view** for faster performance monitoring
- **Global filters** allow drill-down without creating separate reports
- **Analysis flows** from overall performance to regional, yearly, product, profitability, and branch-level views
- **Top 10 Nett Sales by Province** is emphasized as the main visual to highlight geographic contribution
- **Supporting analysis** helps identify sales stability, margin consistency, and branch rating gaps



5. Key Insights & Recommendations

Key Insights

1	West Java contributes **~29.5%** of total Nett Sales.	West Java is the largest sales contributor, showing a strong concentration of revenue in one province.
2	Top 5 provinces contribute **~53.7%** of total Nett Sales.	More than half of total sales come from a small group of provinces, indicating geographic concentration.
3	Annual Nett Sales remained stable at **~Rp80B** per year.	Sales performance was consistent from 2020 to 2023, but there was no significant year-over-year growth.
4	Average Profit Margin remained stable at **~22.6%**	Profitability stayed consistent throughout the period despite relatively flat annual sales.
5	**Some branches have high Branch Ratings but lower Transaction Ratings.**	Top-rated branches recorded Branch Ratings around 4.7–4.8, while Transaction Ratings remained around 4.0, indicating a consistent rating gap.

Recommendations

Maintain performance in West Java and use its sales pattern as a benchmark for other regions.
Explore growth opportunities in lower-contributing provinces to reduce reliance on the top-performing regions.
Review product mix, promotion, and regional strategy to identify opportunities for future sales growth.
Maintain current profitability while monitoring discount levels and product mix.
Investigate customer feedback and transaction processes at branches with the largest rating gaps.

Thank You

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